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Quick reference guide to low tech design for sustainability

Collective Resilience

Maintainability
Can be maintained and repaired by users themselves using standard parts and materials

Design for Repair
How might we design to encourage repair?

Design for continual use
How can we encourage users to repeatedly and consistently use the solution?

Right to Repair
How can we ensure that users feel empowered to repair the design?

Design for modularity
How might we make the solution modular?

Planned Maintenance
How can we ensure that the user understands when and how to maintain the solution?

Repair Cafés
How can we support the community to repair and maintain the solution?

Accessibility
Offers maximum ease of use

Task Analysis
How might we help the user better achieve their task goals?

Stakeholder Analysis
How might we ensure that this solution works for everyone involved?

Adaptability
How can we make sure that the solution is able to be used by different types of users?

Inclusive Design
How can we develop a solution to reduce a mismatch between the solution and the user interactions?

Co-Design
How can we develop the solution with the community?

Labelling
How might we design the solution so that its operation and maintenance are understandable to all?

Tuck behind front cover

Glue Back cover here

Localisation
Made from resources that are exploited and transformed as locally as possible

Production On Demand
How can we design the solution so it can be made only when its required?

Distributed Manufacturing
How can we design the solution so it can be made in multiple locations at once

Local Production
How can we design the solution so that it can be made as close to the user as possible?

Local Material Supply
How can we use materials from the local area?

Cottage Industry
How can we develop this solution to provide local opportunities

Localisation
How can we embed the solution deep within the local community?

Durability
Present maximum technical function, ecological and human viability for the longest period of time

Physically Durable Design
How can we ensure the design doesn't break during use?

Emotionally Durable Design
How can we encourage the user to love the product more?

Design for long life
How might we encourage users to still be using this design in 100 years?

Compatibility
How can you ensure that your design works with other solutions found in its environment?

Upgradability
How could you future proof your solution?

Utility
How can we make the solution as useful as it could possibly be?

Game Guide

Efficiency
Minimise the consumption of energy and resources from production to end of life

Design for disassembly
How can we design this so it comes apart easily?

Design for recyclability
How might we make sure that the design can be recycled?

Minimise material use
How can we use as little material as possible?

Zero Waste
How can we reduce waste to zero?

Lightweight design
How can we make the design as light as we can?

Essentialism
How can we reduce the solution to the bare minimum to function?

Empowerment
Gives power to citizens and communities

Systems thinking
How might we develop better systems around this solution?

Ethical Supply Chain
How might we ensure that we have an ethical supply chain?

Behaviour Change
How might we develop the solution to encourage more sustainable use behaviours?

Product as a Service
What would this solution be like if it was offered as a service?

Design for Resilience
How might we make the solution able to continue to be used with social or cultural developments?

Design for Equity
How can we promote responsible and equal use with the solution?

Cultural Transformation

LT4SUSTAIN is a European Erasmus+ project which brings together 5 universities and an association, and aims to train in the challenges of socio-ecological transition by low-tech

Through the development of hackathons or other such events, aimed at students and industry, we aim to improve inclusivity in the technology ownership and design process.

The objective is to develop scientific and entrepreneurial skills on sustainable and useful solutions using a Low-tech paradigm.

The project partners are:
Université de Technologie de Troyes
Institut Polytechnique de Grenoble
Association Léonard de Vinci
Low-tech Lab Grenoble
Technological University Dublin
Université de Mons

Set a 30 minute timer. In this time brainstorm ideas to solve your users' problem, using the card prompts as guidance.

At the end of the 30 minutes session put your ideas to the side. Discard one Idea Card and draw a single Wild Card and Idea Card from each deck. Choose an Idea Card to discard and select 3 new prompts from each category. Your hand should include 1 Wildcard and 2 Idea Cards.

Start a new 30 minute timer and brainstorm using your new prompts. At the end of the brainstorming discard an Idea Card and as before draw a Wildcard and a new Idea Card.

Continue this process until you only have wild cards in your hand. This is your last ideation round. You should consider your previous ideas with all your wildcard prompts.

As a team consider all the ideas you have created. Discuss them and then use your dot stickers to vote for your favourite idea.

Strong Sustainability

Sobriety
Refocus on the essentials and tend toward the technological optimum - lowest technological intensity and greatest simplicity

Design for minimal energy
How might we design to use as little energy as possible during use?

Minimal viable product
How can we make this the simplest solution it can be?

Standardisation
How can we ensure the solution works with standard products?

Eco sensitive materials
How might we incorporate bio-based materials?

Dematerialisation
How might we reduce the volume of materials used during the manufacture of the solution?

Circular Design
How can you make your solution part of a circular design loop?

Commons
Promotes sharing of knowledge and know-how, cooperation, solidarity and social cohesion

Open Source Design
How can we leverage the power of open source community development?

Vernacular Design
How can we ensure that anyone can learn how to make this?

Customisation
How can we make the solution unique to the user?

Understandable design
How could this solution be implemented without the need for instructions or a manual?

Sharing Economy
How can we encourage the solution to be shared amongst the community?

Maker Community / DIY
How can we develop a maker community to support this solution?

Simplification
Decomplexifies based on needs and vulnerabilities

Design for single material
How can we only use one material in the manufacture of the solution?

Design for Ease of Use
How can we make the solution easy for anyone to use?

Design for Universal Use
How can we ensure that everybody can equitably benefit from this solution?

Design for minimal parts
How can we reduce the number of parts or components in the solution system?

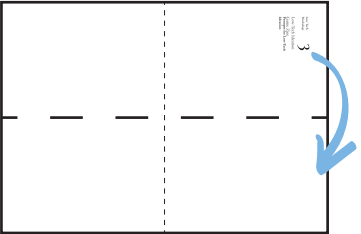
Product Honesty
How can we ensure that materials and systems in the design are visible and unobscured?

Functionalism
How can we make the solution as functional as possible?

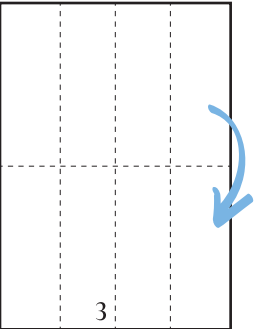


Low-Tech Ideation
Game Zine
Prompts for Low-Tech
Ideation

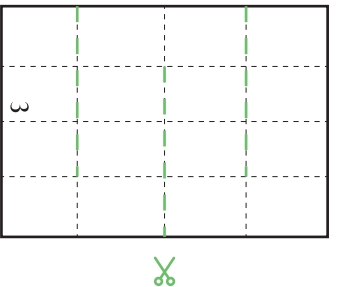
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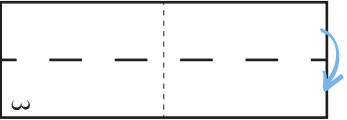
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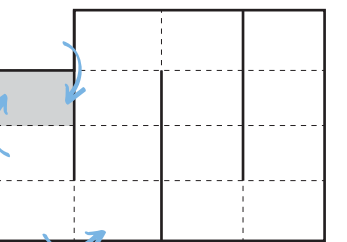
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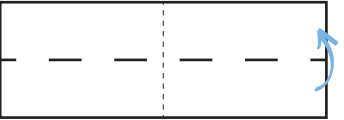
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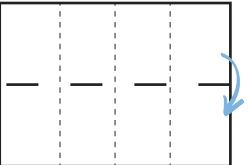
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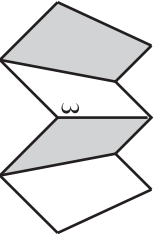
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